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Question Paper Code: 14650

B.E. / B.TECH. DEGREE EXAMINATION, NOVEMBER / DECEMBER 2018

Third Semester

B.E. – Computer Science and Engineering

15CS303 – COMPUTER ORGANIZATION AND ARCHITECTURE

(Regulations: Mepco – R2015)

Duration: 3 Hours Max.: 100 Marks

Answer ALL Questions

PART A – (10 × 2 = 20 Marks)

1. Assume that a program has 100 instructions. 50% of those instructions requires 4 basic steps for execution. 25% requires 5 steps and the remaining 25% requires 6 steps. The clock rate of the processor is 2 GHz. Compute the total program execution time.

2. Convert the following pair of decimal numbers to 5-bit signed 2-s complement binary numbers and add them: 17 and -8. Also, state whether overflow occurs or not in this case.

3. The register that includes the address of the memory unit is termed as the ____.

   A) MAR B) PC C) IR D) MDR

4. The computer architecture aimed at reducing the time of execution of instructions is _____.

   A) CISC B) RISC C) ISA D) Intel

5. Data hazards occurs when _____.

   A) Greater performance loss

   B) Pipeline changes the order of read/write access to operands

   C) Some functional unit is not fully pipelined

   D) Machine size is limited

6. Consider the following sequence of instructions :

   ADD #20, R1

   MUL #3, R2

   ADD R4, R3

   Show the execution of the above instructions in a four-stage pipeline.

7. An address space is specified by 24 bits and the corresponding memory space by 16 bits. How many address bits are there in virtual and main memory?

8. How many memory chips are needed to design 4 M × 16 memory system using 512 K × 8 static memory chips?

9. The high speed version of the SCSI bus use a technique known as _____.

10. In daisy chaining, device 0 will pass signal only if it has:

A) Interrupt request B) No interrupt request

C) Both A and B D) None of the above

PART B – (5 × 16 = 80 Marks)

11. a) i. Registers R1 and R2 contain values 1000 and 2300. What is the effective address of the memory operand in each of the following instructions? Also, identify the addressing mode of each instruction.

Load 130(R1), R5
Move #2000, R5
Store R5, 20(R1, R5)
Add -(R2), R5
Sub (R1)+, R5
Store R1, 45(R5, R2) (8 Marks)

11. a) ii. Perform integer division using any one method on the following:

Dividend: 1010 Divisor: 11. (8 Marks)

OR

11. b) Multiply the following numbers represented in 2's complement using Booth's algorithm.

A = 110110 and B = 0011101

Multiply the above numbers using bit-paring and verify the result of multiplication. (16 Marks)

12. a) i. Compare and contrast between RISC and CISC processor. (6 Marks)

12. a) ii. Illustrate how hardwired control method generates the control signals needed for proper sequencing while executing an instruction with suitable sketches and examples. (10 Marks)

OR

12. b) i. Enumerate the various steps needed for executing an instruction with suitable example. (6 Marks)

12. b) ii. Write the sequence of control steps required for single and multi bus organizations for the instruction :

Add the contents of memory location ADDR to register R1 (10 Marks)

13. a) Consider the following sequence of instructions executed in a pipelined system.

$A \leftarrow B \times 10$

$C \leftarrow A+10$

Does the execution of above instructions cause any hazards? If so, identify it. Provide solutions to address the hazard. (16 Marks)

OR

13. b) i. How the hazards caused by branch instructions are handled? Summarize the approaches with suitable illustrations. (10 Marks)

13. b) ii. Present a brief outline of how the performance of CISC processors can be improved by pipelining. (6 Marks)

14. a) i. Suppose physical addresses are 32 bits wide, there is a cache containing 256 K words of data (not including tag bits), and each cache block contains 4 words. For each of the following cache configurations,

A) direct mapped B) 2-way set associative

C) 4-way set associative D) fully associative

specify how the 32-bit address would be partitioned. For example, for a direct mapped cache, you would need to specify which bits are used to select the cache entry and which bits are used to compare against the tag stored in the cache entry. (10 Marks)

14. a) ii. Compare and contrast between synchronous and asynchronous DRAM memory chips. (6 Marks)

OR

14. b) i. Describe the number of bits required in each entry of a TLB that has the following characteristics:

A) Virtual addresses are 32-bits wide

B) Physical addresses are 31-bits wide

C) The page size is 2K bytes

D) The TLB contains 16 entries of the page table

E) The TLB is direct-mapped (10 Marks)

14. b) ii. Assess the role of DMA in improving the performance of a computer system. Is there any other alternative to DMA? Defend your answer. (6 Marks)

15. a) i. Enumerate the sequence of events involved in handling an interrupt request from a single device. (6 Marks)

15. a) ii. Define bus arbitration. Identify the methods used for arbitration of bus. With suitable sketches, illustrate them. (10 Marks)

OR

15. b) i. Write a detailed note on SCSI and SAS standards. Compare them. (8 Marks)

15. b) ii. Examine the sequence of events that take place when disk read operation is performed using SCSI controller. (8 Marks)